



The metal detector described here is inexpensive and easy-to-build.

Circuit and working

Fig. 1 shows the circuit diagram of the low-cost metal detector. The circuit is built around sensing coil L1*, transistors BC547 (T1 through T3), popular timer NE555 (IC1) in astable mode and a few other components.

Test Points

Test point	Details
TP0	0V
TP1	9V when switch S1 is on
TP2	6.2V when switch S1 is on
TP3	Low when there is no metal near L1* and high when there is metal near L1*

PARTS LIST

Semiconductors:

- IC1 - NE555 timer
- T1-T3 - BC547 npn transistor
- ZD1 - 6.2V zener diode
- D1, D2 - 1N4148 signal diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- | | |
|-------------|---------------------|
| R1, R2, R10 | - 56-kilo-ohm |
| R3 | - 2.7-kilo-ohm |
| R4 | - 3.3-kilo-ohm |
| R5, R9 | - 1-kilo-ohm |
| R6 | - 2.2-kilo-ohm |
| R7 | - 22-kilo-ohm |
| R8 | - 4.7-kilo-ohm |
| VR1 | - 5-kilo-ohm preset |

Capacitors:

- | | |
|--------|---------------------------|
| C1-C3 | - 220pF ceramic disk |
| C4, C7 | - 0.1μF ceramic disk |
| C5, C9 | - 0.01μF ceramic disk |
| C6 | - 100μF, 25V electrolytic |
| C8 | - 0.047μF ceramic disk |
| C10 | - 47μF, 25V electrolytic |

Miscellaneous:

- CON1 - 2-pin connector terminal
- S1 - On/off switch
- LS1 - 8-ohm, 0.5W speaker
- L1* - 20 turns of 26SWG wire over
6cm diameter air core
- 9V regulated power supply
- 8-pin IC base

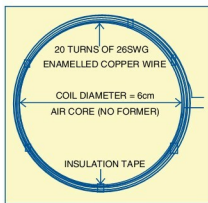


Fig. 2: Details of the sensor coil L1*

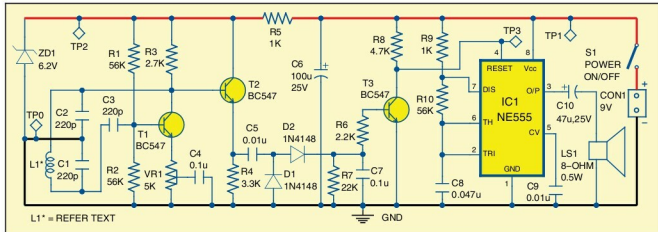


Fig. 1: Circuit diagram of the metal detector

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